

Prescribing anomaly detection

DS2_Task3

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Submitted By: Anurag Chawla

Data Science Intern

1. Introduction

This project focuses on detecting unusual prescribing behaviour across NHS Clinical Commissioning Groups (CCGs) using machine learning-based anomaly detection techniques. The analysis was performed using prescribing data obtained from the Open Prescribing API.

The primary objective was to identify abnormal prescribing patterns that may indicate:

- unusually high prescribing costs,
- excessive prescribing intensity,
- abnormal medication quantities,
- or potentially inefficient prescribing practices.

To achieve this, the Isolation Forest algorithm was applied to engineered prescribing indicators derived from prescribing cost, item counts, quantity information, and population-adjusted prescribing rates.

2. Objective

The main objectives of this project were:

- To analyse NHS prescribing data at the CCG level.
- To engineer normalized prescribing indicators for meaningful comparison.
- To apply the Isolation Forest algorithm for unsupervised anomaly detection.
- To identify and flag unusual prescribing behaviors.
- To generate anomaly outputs for further healthcare analysis.

3. Dataset Description

The datasets used in this project were obtained from the OpenPrescribing API, an NHS healthcare prescribing platform that provides publicly accessible prescribing statistics for Clinical Commissioning Groups (CCGs) across the United Kingdom.

To perform prescribing anomaly detection at the CCG level, two separate datasets were collected and merged.

3.1. Spending by CCG Dataset (spending-by-ccg.csv)

The first dataset contained prescribing expenditure and prescribing activity information for NHS CCGs.

Dataset Dimensions

- Rows: 6,421
- Columns: 6

```
df1.shape
(6421, 6)
```

Major Variables Included

```
df1.head()
```

	actual_cost	date	items	quantity	row_id	row_name
0	4894995.14	2021-03-01	738312	55076671.6	00L	NHS NORTHUMBERLAND
1	2558875.09	2021-03-01	376146	27858759.8	00N	NHS SOUTH TYNESIDE
2	4402168.34	2021-03-01	773348	46757988.3	00P	NHS SUNDERLAND
3	2394119.19	2021-03-01	310561	23321370.6	00Q	NHS BLACKBURN WITH DARWEN
4	2903234.99	2021-03-01	414514	27514848.1	00R	NHS BLACKPOOL

Purpose of the Dataset

This dataset provided the core prescribing activity required for anomaly detection analysis. It helped identify:

- unusually high prescribing expenditure,
- abnormal prescribing volume,
- and irregular medicine quantity patterns across NHS regions.

3.2. Organization Details Dataset (org_details.csv)

The second dataset contained NHS organizational and patient population information for each CCG.

Dataset Dimensions

- Rows: 6,360
- Columns: 4

```
df2.shape
(6360, 4)
```

Major Variables Included

df2.head()

	date	row_id	row_name	total_list_size
0	2021-03-01	00L	NHS NORTHUMBERLAND	330624
1	2021-03-01	00N	NHS SOUTH TYNESIDE	158311
2	2021-03-01	00P	NHS SUNDERLAND	284234
3	2021-03-01	00Q	NHS BLACKBURN WITH DARWEN	179447
4	2021-03-01	00R	NHS BLACKPOOL	174961

Next steps: [New interactive sheet](#)

Purpose of the Dataset

This dataset was important because raw prescribing totals alone can be misleading. Larger CCGs naturally generate higher prescribing costs and quantities due to larger patient populations.

The population information allowed prescribing activity to be normalized for fair comparison across all NHS regions.

3.3. Dataset Merging

Both datasets were merged using:

- row_id
- date

This ensured that prescribing activity matched correctly with the corresponding NHS population data for the same CCG and reporting month.

After merging and preprocessing, the final analytical dataset contained:

Merged shape: (6360, 7)

	actual_cost	date	items	quantity	row_id	row_name	total_list_size
0	4894995.14	2021-03-01	738312	55076671.6	00L	NHS NORTHUMBERLAND	330624
1	2558875.09	2021-03-01	376146	27858759.8	00N	NHS SOUTH TYNESIDE	158311
2	4402168.34	2021-03-01	773348	46757988.3	00P	NHS SUNDERLAND	284234
3	2394119.19	2021-03-01	310561	23321370.6	00Q	NHS BLACKBURN WITH DARWEN	179447
4	2903234.99	2021-03-01	414514	27514848.1	00R	NHS BLACKPOOL	174961

4. Feature Engineering

To improve anomaly detection quality, several healthcare-oriented prescribing indicators were engineered from the merged dataset.

Engineered Features Used

- `cost_per_patient`
 - Average prescribing cost per patient.
- `items_per_patient`
 - Average prescribed items per patient.
- `quantity_per_patient`
 - Average medicine quantity prescribed per patient.
- `cost_per_item`
 - Average prescribing cost per prescription item.
- `quantity_per_item`
 - Average quantity associated with each prescribed item.

Features created!

	actual_cost	date	items	quantity	row_id	row_name	total_list_size	cost_per_patient	items_per_patient	quantity_per_patient	cost_per_item	quantity_per_item
0	4894995.14	2021-03-01	738312	55076671.6	00L	NHS NORTHUMBERLAND	330624	14.805323	2.233087	166.584010	6.629982	74.598099
1	2558875.09	2021-03-01	376146	27858759.8	00N	NHS SOUTH TYNESIDE	158311	16.163596	2.375994	175.974884	6.802877	74.063688
2	4402168.34	2021-03-01	773348	46757988.3	00P	NHS SUNDERLAND	284234	15.487832	2.720815	164.505261	5.692351	60.461769
3	2394119.19	2021-03-01	310561	23321370.6	00Q	NHS BLACKBURN WITH DARWEN	179447	13.341651	1.730656	129.962444	7.709014	75.094331
4	2903234.99	2021-03-01	414514	27514848.1	00R	NHS BLACKPOOL	174961	16.593612	2.369179	157.262751	7.003949	66.378574

4.1. Importance of Engineered Features

The engineered prescribing indicators helped:

- reduce population-size bias,
- normalize prescribing behavior,
- identify unusually expensive prescribing patterns,
- detect excessive prescribing intensity,
- and improve the effectiveness of the Isolation Forest anomaly detection model.

4.2. Shape after feature engineering

```
df.shape
(6360, 12)
```

4.3. Feature Selection

After preprocessing and feature engineering, a feature selection step was performed to identify the most relevant variables for anomaly detection.

Instead of using raw prescribing totals directly, normalized prescribing indicators were selected because they provide a more accurate representation of prescribing behavior across NHS CCG regions with different population sizes.

The following features were selected for model training:

Selected Feature	Description
cost_per_patient	Average prescribing cost per patient
items_per_patient	Average prescribed items per patient
quantity_per_patient	Average medicine quantity prescribed per patient
cost_per_item	Average prescribing cost per prescription item
quantity_per_item	Average quantity associated with each prescribed item

Reason for Selecting These Features

- normalize prescribing behavior across regions,
- highlight abnormal prescribing intensity,
- identify unusually expensive prescriptions,
- improve the effectiveness of anomaly detection.

Using normalized healthcare indicators allowed the Isolation Forest algorithm to detect genuinely unusual prescribing patterns rather than simply identifying larger NHS regions with naturally higher prescribing volumes.

Missing values were handled during preprocessing, and all numerical features were standardized using Standard Scaler before model training.

5. Isolation Forest Model

The Isolation Forest algorithm was selected because it is highly effective for unsupervised anomaly detection in high-dimensional healthcare datasets.

The model works by randomly isolating observations in the feature space. Records that are isolated more quickly are considered anomalous.

The model configuration used was:

```
model = IsolationForest(  
    n_estimators=100,  
    contamination=0.05,  
    random_state=42  
)  
model.fit(X_scaled)
```

IsolationForest

IsolationForest(contamination=0.05, random_state=42)

Where:

- `n_estimators=100` defines the number of isolation trees,
- `contamination=0.05` assumes approximately 5% anomalies in the dataset,
- `random_state=42` ensures reproducibility.

6. Results and Findings

The Isolation Forest model successfully identified unusual prescribing patterns across NHS CCGs.

Anomaly Detection Summary

```
... Isolation Forest applied!  
  
Anomaly counts:  
anomaly_flag  
Normal      6042  
Anomaly      318  
Name: count, dtype: int64
```

The detected anomalies represented approximately: $\frac{318}{6360} \times 100 \approx 5.0\%$

of the complete dataset, which aligned with the configured contamination parameter.

7. Interpretation of Anomalies

The anomalous prescribing records generally exhibited one or more of the following characteristics:

- unusually high prescribing cost per patient,
- excessive prescribing quantity,
- abnormal prescribing intensity,
- unusually high cost per prescribed item,
- irregular medication quantity behavior.

These patterns may indicate:

- inefficient prescribing behavior,
- unusual healthcare demand,
- data irregularities,
- or prescribing practices requiring further investigation.

8. Visualization & Exploratory Analysis

The visual analysis provided strong evidence that the Isolation Forest model successfully identified unusual prescribing patterns across NHS regions.

8.1 Isolation Forest Anomaly Score Distribution

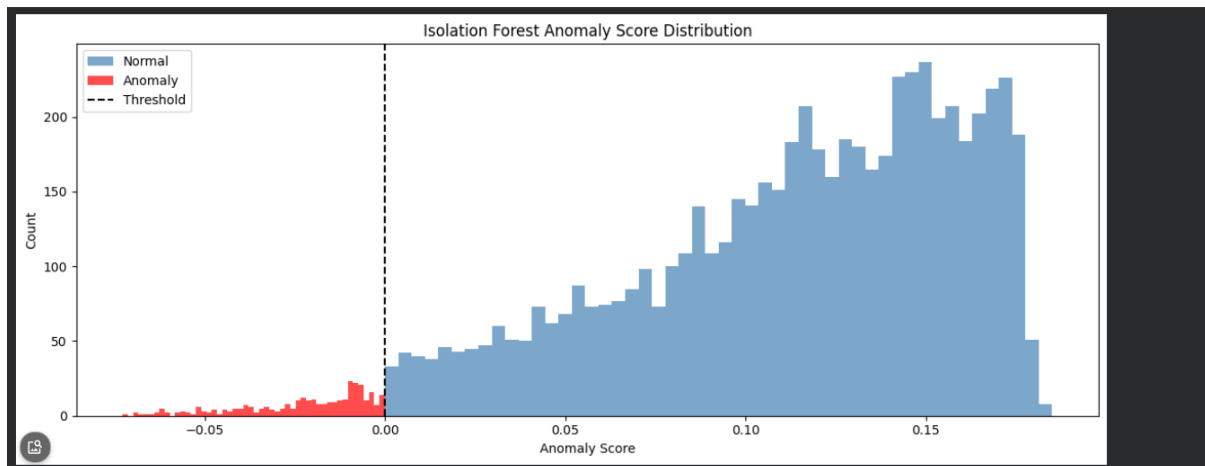
A histogram-based anomaly score distribution plot was created to visualize how the Isolation Forest model separated normal prescribing behavior from anomalous prescribing records.

The visualization displayed:

- anomaly scores for normal observations,
- anomaly scores for anomalous observations,
- and the anomaly threshold boundary.

Key Observations

- Most normal prescribing records were concentrated on the positive side of the anomaly score distribution.
- Anomalous records were primarily located in the negative score region.
- The threshold line at score 0 clearly separated normal and anomalous prescribing behavior.
- The visualization confirmed that the model effectively isolated unusual prescribing patterns from standard NHS prescribing activity.



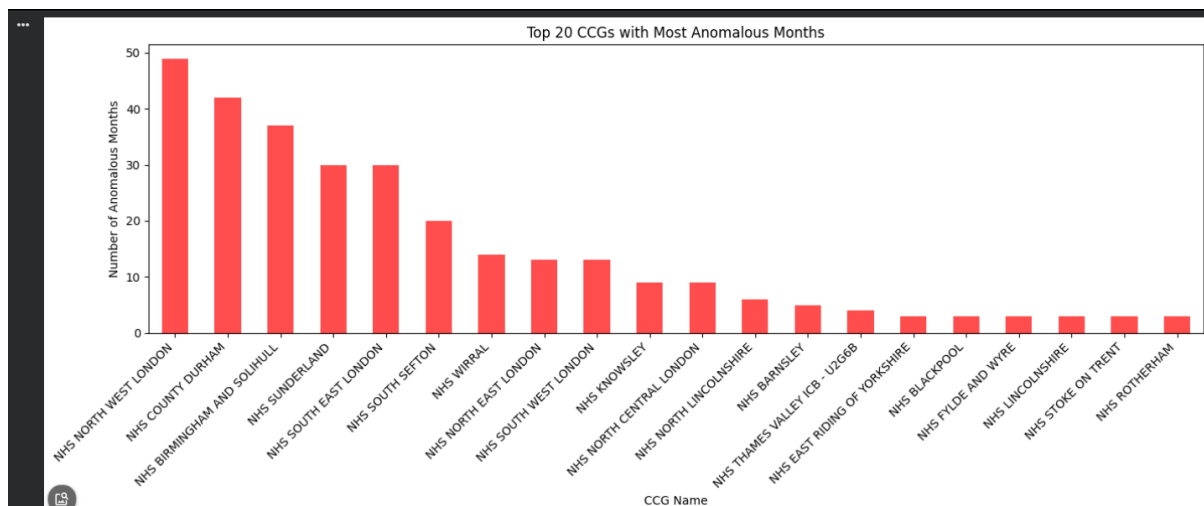
8.2 Top 20 CCGs with Most Anomalous Months

A bar chart visualization was generated to identify NHS CCG regions that exhibited the highest number of anomalous prescribing months.

The graph ranked the top 20 CCGs based on the frequency of detected anomalies.

Key Observations

- Certain NHS regions consistently exhibited unusual prescribing behavior across multiple months.
- Regions such as:
 - NHS North West London,
 - NHS County Durham,
 - NHS Birmingham and Solihull,
 - and NHS Sunderland
 showed comparatively higher anomaly counts.
- These repeated anomalies may indicate:
 - unusually high prescribing intensity,
 - irregular medicine expenditure,
 - or persistent prescribing inefficiencies.



8.3 Anomaly Score Heatmap Across Time

A heatmap visualization was created to analyze anomaly score behavior over time for the top 10 most anomalous NHS CCGs.

The heatmap represented:

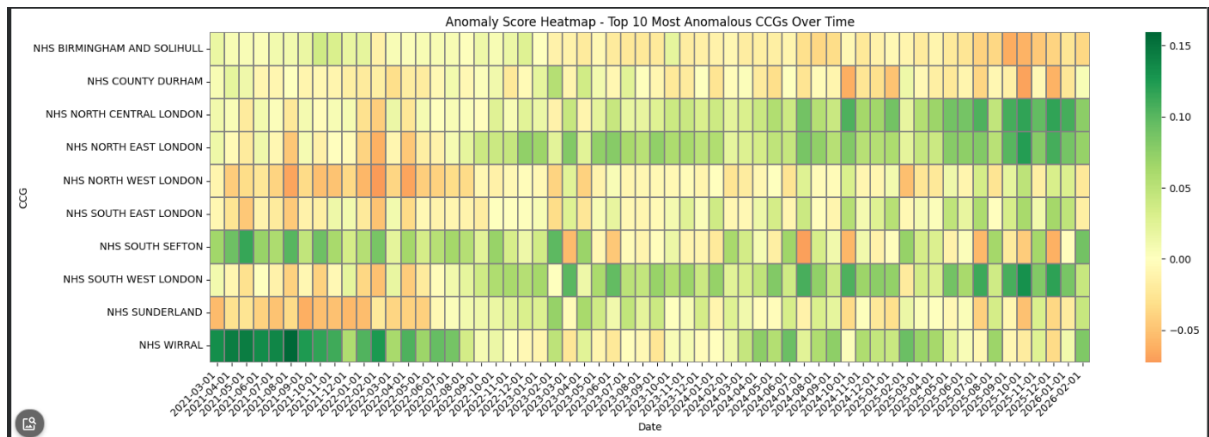
- NHS CCG regions on the vertical axis,
- reporting dates on the horizontal axis,
- and anomaly scores through color intensity.

Heatmap Interpretation

- Red regions represented stronger anomalous behavior.
- Green regions represented more normal prescribing activity.
- Variations in color intensity highlighted changes in prescribing irregularities across different months.

Key Observations

- Several NHS regions displayed persistent anomaly patterns across multiple reporting periods.
- Some CCGs exhibited sudden spikes in anomalous prescribing behavior during specific months.
- Temporal fluctuations suggested variations in prescribing intensity and healthcare expenditure over time.



8.4 Overall Visualization Insights

The combination of:

- anomaly score distribution analysis,
- regional anomaly frequency visualization,
- and temporal anomaly heatmaps

provided a comprehensive understanding of prescribing irregularities within the NHS prescribing dataset.

9. Exporting Anomaly Detection Results

After completing anomaly detection using the Isolation Forest algorithm, the final processed dataset was exported into a structured CSV file for further healthcare analysis and reporting.

The exported file contained:

- original prescribing information,
- engineered healthcare indicators,
- anomaly scores,
- and anomaly classification labels.

Anomaly_flags.csv exported successfully!
Total records: 6368
Anomalies flagged: 318

Preview:

row_id	row_name	date	actual_cost	items	quantity	total_list_size	cost_per_patient	items_per_patient	quantity_per_patient	cost_per_item	quantity_per_item	anomaly_score	anomaly_flag
1270	W2U3Z NHS NORTH WEST LONDON	2022-02-01	19423181.68	2586988	2.059585e+08	2757592	7.043530	0.938133	74.687794	7.5088029	79.613227	-0.072711	Anomaly
4257	01T NHS SOUTH SEFTON	2024-07-01	3210821.27	381590	3.672543e+07	160286	20.031826	2.380682	229.124365	8.414322	96.243162	-0.069772	Anomaly
5916	84H NHS COUNTY DURHAM	2025-10-01	11498210.26	1649407	9.261526e+07	570736	20.146285	2.889965	162.273385	6.971118	56.150642	-0.068588	Anomaly
634	W2U3Z NHS NORTH WEST LONDON	2021-08-01	19860488.71	2627320	2.101422e+08	2702283	7.349522	0.972259	77.764687	7.559220	79.983478	-0.067370	Anomaly
1482	W2U3Z NHS NORTH WEST LONDON	2022-04-01	20599286.80	2743111	2.204795e+08	2771172	7.433421	0.989874	79.561821	7.509462	80.375709	-0.066780	Anomaly
5913	71E NHS LINCOLNSHIRE	2025-10-01	17459326.54	2275636	1.348757e+08	826065	21.135536	2.754790	163.274970	7.672284	59.269470	-0.064978	Anomaly
5794	15E NHS BIRMINGHAM AND SOLIHULL	2025-09-01	23599550.31	2324214	2.849935e+08	1647326	14.325975	1.410901	173.003732	10.153777	122.619322	-0.063394	Anomaly
6057	01J NHS KNOWSLEY	2025-12-01	3503463.95	430767	4.012745e+07	174360	20.093278	2.470561	230.141364	8.133083	93.153487	-0.062958	Anomaly
5527	01J NHS KNOWSLEY	2025-07-01	3370545.51	415526	4.017620e+07	174078	19.362272	2.387010	230.794262	8.111515	96.687580	-0.062301	Anomaly
638	00P NHS SUNDERLAND	2021-09-01	4417484.07	770978	4.535975e+07	284687	15.516986	2.708160	159.332007	5.729715	58.834041	-0.062295	Anomaly

10. Limitations

1. Cloudflare API Restriction

The OpenPrescribing API could not be accessed programmatically due to Cloudflare bot protection data had to be downloaded manually via browser, which limits automation and scalability of the pipeline.

2. Only 2 Datasets Used

The analysis used only total spending and list size data. Drug-category level data (antibiotics, opioids, antidepressants etc.) was available but not included richer drug-level features could improve anomaly detection quality.

3. CCG Restructuring

Some NHS CCGs merged or reorganised during 2021-2026. This could cause artificial anomalies where a CCG's list size or spending changes abruptly due to boundary changes, not genuine prescribing behaviour.

11. Deliverables

The following outputs were successfully generated:

[anomaly_flags.csv](#)

[Jupyter Notebook](#)

12. Conclusion

This project successfully demonstrated the use of Isolation Forest for healthcare prescribing anomaly detection using NHS OpenPrescribing data.

By engineering population-adjusted prescribing indicators and applying unsupervised machine learning techniques, the analysis was able to identify unusual prescribing behaviors across NHS CCGs.

The generated anomaly flags and visualizations provide valuable insights for:

- healthcare monitoring,
- prescribing audits,
- resource optimization,
- and further healthcare analytics investigations.

Overall, the project highlights the effectiveness of anomaly detection techniques in identifying abnormal healthcare prescribing patterns in large-scale real-world datasets.